



RMC.BestFit 2.0 Verification Report

Numerical verification, independent package parity,
published benchmarks, and scientific finding traceability

RISK MANAGEMENT CENTER

Verification program draft
RMC.Numerics 2.1.4 dependency
24 July 2026



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This report is generated from the living verification Markdown. A claim is verified only when its focused test, independent oracle, tolerance, and result are recorded.

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CHAPTER 01

RMC.BestFit 2.0 Verification Report

This is the living formal verification and validation record for `RMC.BestFit.Verification`. It separates scientific numerical evidence from fast unit and regression coverage.

Current status

Program area	Status	Evidence
Repository integration	Validated 24 July 2026	Debug Verification rebuild: 0 warnings/errors; Release solution log: 0 Verification project references
Public API baseline	Captured	Exact exported type/member/enum baseline enforced by <code>PublicApiCompatibilityTests</code>
Fast regression gate	Passed	Core 3,028; UI 564; App 427; 0 failed and 0 skipped
External environments	Locked	R 4.4.3 with 131 packages; Python with 15 packages
Test ownership audit	In progress	Initial duplicate removals and <code>FittingAnalysis</code> split are recorded in Test Inventory
Distribution fitting	Paused at defect off-ramp	TR-001 and TR-010 are confirmed; focused post-fix evidence remains pending in Distribution Fitting
Model estimation and diagnostics	Planned after off-ramp	Model Estimation
Bulletin 17C and later phases	Planned	Scientific Review Findings

Evidence rule

A passing build, optimizer convergence, or reproduction of an internal result is not verification. A claim is verified only when its test has an analytical, independently implemented, external-package, or published-result oracle with declared provenance and tolerance.

Execution rule

The full Verification project is never run as one suite. Each result in this report is produced by exactly one fully qualified method through:

```
.\scripts\run-verification-test.ps1 -Test Namespace.Class.Method
```

The runner rejects broad filters and requires exactly one TRX result.

Traceability states

- **Planned:** no executable scientific test yet.
- **Ready - user run:** test compiles and an exact command is available.
- **Passed:** the focused TRX and oracle artifact have been reviewed.
- **Failed:** the focused test contradicted the claim or exposed a defect.
- **Blocked:** an external dependency, upstream correction, or scientific decision prevents completion.

Methodology | References

CHAPTER 02

Verification Methodology

Purpose

The program establishes claim-specific evidence for statistical correctness, numerical stability, and external parity. Unit tests remain in `RMC.BestFit.Tests`; this report covers only numerical verification and validation.

Evidence hierarchy

1. Closed-form analytical solution.
2. Independently implemented algorithm.
3. Trusted external package with a documented parameterization crosswalk.
4. Published worked example or table.
5. Simulation coverage with a predeclared design and Monte Carlo acceptance interval.

Regression pins and estimator smoke tests are useful engineering checks but are not independent verification.

Reproducibility contract

Each external artifact records the source dataset, package and runtime versions, generator path, command, seed, tolerance and rationale, SHA-256 hash, generation date, and affected finding IDs. R and Python generate committed CSV or JSON files. C# tests do not invoke those runtimes.

Tolerance policy

Comparison	Default
Deterministic analytical arithmetic	Absolute tolerance approximately 10^{-10}
Cross-language log likelihood and criteria	Absolute 10^{-8} , relative 10^{-7}
Optimizer parameter parity	Scaled tolerance no looser than 10^{-5} , plus objective parity
PSIS aggregate values	10^{-6}
Pareto k	10^{-4}
Coverage	Nominal rate must lie in the predeclared binomial interval

Exceptions must be documented before the C# result is observed.

Defect off-ramp

When independent evidence confirms a defect, scientific progression stops. The finding is updated and a focused correction plan is presented. Production correction begins only after approval and must preserve public signatures and serialization whenever possible.

After a correction, the three fast test projects, XML documentation gate, public API compatibility check, focused verification test, technical-reference chapter, finding entry, and this report must all agree before the finding is marked fixed.

CHAPTER 03

Verification Test Inventory

Ownership rule

Fast behavior, validation, serialization, property, state, event, exception, and regression tests belong in `RMC.BestFit.Tests`. Numerical verification remains only when a test uses an analytical, external, published, independently implemented, recovery, or coverage oracle.

Initial migration - 24 July 2026

Verification source	Disposition	Fast-test destination or coverage
<code>ModelEstimation/InfluenceDiagnosticsTests.cs</code>	Removed as unit/DTO coverage	Existing <code>Diagnostics/InfluenceDiagnosticsTests.cs</code>
<code>ModelEstimation/PointwiseLogLikelihoodTests.cs</code>	Removed as decomposition/unit coverage	Existing univariate, bivariate, time-series, mixture, point-process, and spatial unit tests
<code>ModelEstimation/PredictiveChecksTests.cs</code>	Removed as unit/regression coverage	Existing expanded prior/posterior predictive and result DTO unit tests
<code>DistributionFitting/FittingAnalysisTests.cs</code>	Split	Deterministic state/event/regression cases moved to <code>FittingAnalysisRegressionTests</code> ; published-data comparisons retained

Four legacy assertions were not retained: a probability-ordinate mutation test contradicted the established no-refit contract; generic large-sample and distribution-success counts had no oracle; and the outlier smoke test reproduced TR-010 (IsEstimated true with zero successful candidates). TR-010 is recorded as a confirmed defect, and its post-fix regression will be added with the approved correction.

Remaining audit

The following mixed files require method-level splitting before their domain phase begins:

Area	Mixed files
Model estimation	<code>GeneralizedMethodOfMomentsTests.cs</code> , <code>MaximumAPosterioriTests.cs</code>
Rating curve	<code>RatingCurveTests.cs</code> , <code>RatingCurveAnalysisTests.cs</code>
Time series	Model and analysis test files containing both constructors/properties and estimator recovery
Bivariate and spatial	Files containing both DTO/state checks and fitted-model recovery

The audit is intentionally marked **in progress**. No remaining mixed file is represented as verification-grade until its methods have been classified and moved.

CHAPTER 04

Distribution Fitting Verification

Scope

The first phase verifies the 15 supported univariate families, the fitting pipeline, goodness-of-fit metrics, and findings TR-001, TR-002, TR-009, and TR-010.

Distribution matrix

Family	Primary oracle	Secondary oracle	Status
Normal	Closed-form MLE	SciPy	Planned
Log10-Normal	Closed-form transformed MLE	SciPy	Planned
Ln-Normal	Closed-form transformed MLE	SciPy	Planned
Exponential	Closed form / published data	SciPy	Planned
Gamma	Imomco	SciPy	Planned
GEV	Imomco	SciPy	Planned
Generalized Logistic	Imomco	Published hydrology example	Planned
Generalized Normal	Imomco	Published hydrology example	Planned
Generalized Pareto	Imomco	SciPy	Planned
Gumbel	Imomco	SciPy	Planned
Kappa Four	Analytical branch identities	SciPy and Imomco	Active
Logistic	Closed form / published data	SciPy	Planned
Log-Pearson III	Transformed Pearson III	Imomco	Planned
Pearson III	Imomco	SciPy	Planned
Weibull	Imomco	SciPy	Planned

TR-001 - Kappa Four zero primary shape

For $\kappa = 0$, $h \neq 0$, $z = (x - \xi)/\alpha$, the implemented CDF is

$$F(x) = [1 - h \exp(-z)]^{1/h}.$$

Independent differentiation and inversion give

$$f(x) = \frac{\exp(-z)}{\alpha} [1 - h \exp(-z)]^{1/h-1}$$

and

$$F^{-1}(p) = \xi - \alpha \log \left(\frac{1 - p^h}{h} \right).$$

The focused analytical tests are:

- `RMC.BestFit.Verification.DistributionFitting.KappaFourZeroShapeVerificationTests.ZeroShapeNonzeroHondo_PdfMatchesAnalyticalDerivative`
- `RMC.BestFit.Verification.DistributionFitting.KappaFourZeroShapeVerificationTests.ZeroShapeNonzeroHondo_QuantileInvertsCdf`

Disposition: Confirmed defect by algebra, source review, and the existing compiled probe.

Implementation: External dependency - RMC.Numerics correction requires a focused plan.

Verification: Ready - user run; no pass is claimed.

TR-010 - all-candidate failure reports overall success

A deterministic outlier fixture completed with `IsEstimated == true` while all 15 `FittedDistribution.FitSucceeded` values were false. Source inspection confirms that the outer run sets `IsEstimated = true` whenever the parallel loop itself completes, without requiring a successful candidate.

Disposition: Confirmed defect by deterministic behavioral reproduction and source inspection.

Implementation: Not started; an API-compatible state and status plan is required.

Verification: Post-fix regression and focused verification are pending.

Log10-Normal analytical design

For $x_i = \log_{10}(y_i)$,

$$\hat{\mu} = \frac{1}{n} \sum_i x_i, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \hat{\mu})^2.$$

The test will compare parameters, original-measure data log likelihood including the base-10 Jacobian, CDF values, and quantiles before external optimizer comparisons are accepted.

CHAPTER 05

Model Estimation and Diagnostics Verification

Status

Planned after the TR-001 off-ramp is resolved.

Log10-Normal prior and penalty experiment

A deterministic symmetric log10 dataset and an outlier variant will compare MLE, flat-prior MAP, and Bulletin 17C GMM. A Gaussian prior on μ is mapped to the quadratic penalty using the implementation convention $MSE = \tau_{\mu}^2/n$, subject to the TR-033 objective-scale correction.

Regime	Prior scale and center	Expected location effect	Expected variance effect
Wide centered	$100SE_L$, centered at MLE	None	Negligible
Narrow centered	$0.1SE_L$, centered at MLE	None	Large contraction
Narrow shifted	$0.1SE_L$, shifted by $2SE_L$	Material shift	Large contraction

Exact leave-one-out refits provide the observation-influence oracle. Full-Hessian and deletion covariance comparisons provide the leverage oracle; no sum-to- p assertion is accepted without a derivation.

External model-comparison oracles

- R `loo` 2.10.0 for WAIC and PSIS-LOO.
- R `BayesianTools` 0.1.9 for the selected DIC convention.
- R `gmm` 1.9-1 for estimates, covariance, and Hansen's J .
- R `bbmle` 1.0.25.1 for nuisance-reoptimized profiles.
- Python `ArviZ` 1.2.0 as a secondary WAIC/LOO comparison.

The fixtures will use committed deterministic posterior draws and pointwise log-likelihood matrices so package parity is not confounded with sampler randomness.

CHAPTER 06

Verification References

- Asquith, W. H. `lmomco`: L-moments and hydrologic distributions, CRAN package 2.5.7.
- Bolker, B. and R Development Core Team. `bbmle`: tools for general maximum likelihood estimation, CRAN package 1.0.25.1.
- Chausse, P. `gmm`: generalized method of moments and generalized empirical likelihood, CRAN package 1.9-1.
- Hartig, F. et al. `BayesianTools`: general-purpose MCMC, SMC, and Bayesian diagnostics, CRAN package 0.1.9.
- Vehtari, A. et al. `loo`: efficient leave-one-out cross-validation and WAIC, CRAN package 2.10.0.
- SciPy community. `scipy.stats` reference, version 1.17.1.
- RMC-BestFit version 1 verification report, [docs/reports/RMC-TR-2020-02 - Verification of the Bayesian Estimation and Fitting Software.pdf](#).
- RMC.BestFit 2.0 Technical Reference and Scientific Review Findings.